sees a reflection when some longitudes of Titan are turned toward Earth, and not at other longitudes. All right, you might say, so Titan has oceans and continents, and it was a continent that reflected the signals back to Earth. But if Titan is in this respect like the Earth—for some meridians (through Europe and Africa, say) mainly continent, and for others (through the central Pacific, say) mainly ocean—then we must confront another problem:

The orbit of Titan around Saturn is not a perfect circle. It's noticeably squashed out, or elliptical. If Titan has extensive oceans, though, the giant planet Saturn around which it orbits will raise substantial tides on Titan, and the resulting tidal friction will circularize Titan's orbit in much less than the age of the Solar System. In a 1982 scientific paper called "The Tide in the Seas of Titan," Stanley Dermott, now at the University of Florida, and I argued that for this reason Titan must be either an all-ocean or an all-land world. Otherwise the tidal friction in places where the ocean is shallow would have taken its toll. Lakes and islands might be permitted, but anything more and Titan would have a very different orbit than the one we see.

We have, then, three scientific arguments—one concluding that this world is almost entirely covered with hydrocarbon oceans, another that it's a mix of continents and oceans, and a third requiring us to choose, counseling that Titan can't have extensive oceans and extensive continents at the same time. It will be interesting to see what the answer turns out to be.

What I've just told you is a kind of scientific progress report. Tomorrow there might be a new finding that clears up these mysteries and contradictions. Maybe there's something wrong with Muhleman's radar results, although it's hard to see what it might be: His system tells him he's seeing Titan when it's nearest, when he ought to be seeing Titan. Maybe there's something wrong with Dermott's and my calculation about the tidal evolution of the orbit of Titan, but no one has been able to find any errors so far. And it's hard to see how ethane can avoid condensing out at the surface of Titan. Maybe, despite the low temperatures, over billions of years there's been a change in the chemistry; maybe some combination of